MIL-P-63223A(AR)
5 December 1984
SUPERSEDING
MIL-P-63223(AR)
17 August 1979

MILITARY SPECIFICATION

PROJECTILE, GUIDED, 155MM, M712

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture and acceptance of the M712, 155MM, Guided Projectile, Prime item.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified the following specifications, standards, and handbook of the issue listed in that issue of the Department of Defense Index of Specifications (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-63224 - Projectile, Guided, 155MM, M712, Guidance Section for
MIL-P-63225 - Projectile, Guided, 155MM, M712, Control Section for
MIL-P-63226 - Projectile, Guided, 155MM, M712, Warhead for
MIL-P-63227 - Projectile, Guided, 155MM, M712, Packaging for

FSC: 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-100	-	Engineering Drawing Practices
MIL-STD-105	-	Sampling Procedures and Tables from Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-810	-	Environmental Test Methods
MIL-STD-1171	-	Acceptance and Description Sheets.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US Army Armament Research and Development Center (ARDC)

9300440	- Container, Ammunition, Metal, PA101, for projectile, 155MM, HE, Guided, Cannon Launched, M712, and Training, M823.
9305300	- Projectile, Guided M712
9305335	- Packing assembly, for Projectile, 155MM, HE, Guided, Cannon Launched, M712, and Training M823 with Packing and marking instructions.

US Army Defense Ammunition Center and School

19-48-4159-20PM-1003	- Unitization Procedures for 155MM Copperhead Separate Loading Projectiles, Packed One Per Metal Container, Unitized 6 Containers Per Special 4-way Entry Pallet.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring tivity or as directed by the contracting officer).

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Item definition. The Projectile, Guided, 155MM, M712, hereinafter referred to as the projectile, consists of three sections: guidance section (MIL-P-63224), control section (MIL-P-63225), and warhead section (MIL-P-63226). A container (MIL-P-63227) is used to environmentally protect each projectile during storage and transportation. Launched from a 155 MM howitzer, the projectile receives guidance by acceptance of coded laser signals reflected from a target into an optical seeker system. These signals are translated into appropriate fin commands for projectile proportional guidance to the laser-illuminated target. Upon impact, direct impact and shock wave sensors produce ordnance detonate signals which initiate the explosive train and detonate the shaped charge warhead. The projectile has two flight modes; ballistic and glide. For short to medium range missions a ballistic trajectory is employed. For long range missions (glide), the extended range is attained by gliding after an initial ballistic flight mode.

3.2 Design and construction.

3.2.1 Production drawings. - The projectile shall be fabricated and assembled in accordance with drawings, parts lists, and other documents listed on drawing 9305300.

3.3 Characteristics. - Projectile characteristics shall be as specified herein.

3.3.1 Performance.

3.3.1.1 Warhead section/control section cable resistance. With the warhead section mechanically and electrically assembled to the control section (warhead section cables not connected to guidance section cables) the direct current (DC) resistance characteristics of the warhead section/control section assembly wiring (as measured from the forward end of the warhead section) shall be as shown in Table I and in drawing 9308000. CAUTION: Measuring devices shall provide no more than 1.0 milliamperes (ma) of current to the circuit under test. Test lead resistance is not included in the tabulated values.

3.3.1.2 Warhead integrity. The warhead fuze shall remain in the "SAFE" condition following exposure to the environments of 3.3.2.1 and 3.3.2.2

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TABLE I Connections to Warhead Section

FROM*		TO*		RESISTANCE (OHMS)
FIRE DET 1	3P2-1	CASE		20M
EAI SQ DR	3P2-2	CASE		20M
SQ GND 1	3P2-3	CASE		20M
CHG DET CAP 1	3P2-4	CASE		20M
FUZE GND 1	3P2-5	CASE		20M
FIRE DET 2	3P1-1	CASE		20M
EA 2 SQ DR	3P1-2	CASE		20M
SQ GND 2	3P1-3	CASE		20M
CHG DET CAP 2	3P1-4	CASE		20M
FUZE GND 2	3P1-5	CASE		20M
FIRE DET 1	3P2-1	FUZE GND 1	3P2-5	360K ± 40K
FUZE GND 1	3P2-5	FIRE DET 1	3P2-1	70K ± 20K
EA1 SQ DR	3P2-2	FUZE GND 1	3P2-5	8.1 ± 4.1
SQ GND 1	3P2-3	FUZE GND 1	3P2-5	±1.0
CHG DET CAP 1	3P2-4	FUZE GND 1	3P2-5	900K**
FIRE DET 2	3P1-1	FUZE GND 2	3P1-5	360K ± 40K
FUZE GND 2	3P1-5	FIRE DET 2	3P1-1	70K ± 20K
EA2 SQ DR	3P1-2	FUZE GND 2	3P1-5	8.1 ± 4.1
SQ GND 2	3P1-3	FUZE GND 2	3P1-5	±1.0
CHG DET CAP 2	3P1-4	FUZE GND 2	3P1-5	900K**
SGG SQ DR	3P1-25	SGG SQ	3P1-24	0.3
SGS SQ DR	3P2-25	SGS SQ	3P2-24	0.3
SGG SQ DR	3P1-25	CASE		1M

* The test equipment lead with the positive potential shall be connected to the "from connection. The test equipment shall be set for the minumum resistance range compatible with the maximum expected resistance.

** After 30 seconds.

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3.3.1.3 Functional projectile. A functional projectile is one which, when fired within the conditions outlined in Tables II & IV, shall reach the target area and intersect the target within a circle, in a plane perpendicular to the projectile velocity vector, having a radius of 5.0 feet from designator spot centroid, and detonate with a high order explosion upon impact. The designator spot centroid is defined as the centroid of laser spot energy averaged over the last 2 seconds of projectile flight. The 3σ variation of the spot center during these last 2 seconds shall be held to a one foot diameter circle.

3.3.2 Environmental conditions. The projectile shall be capable of meeting the performance requirements of 3.3.1.2 and 3.3.1.3 after exposure to the environmental requirements of 3.3.2.1 and 3.3.2.2. During exposure to storage and transportation environments, the projectile shall be housed in its container as defined by MIL-P-63227. After exposure to these environmental conditions, the projectile shall show no evidence of physical damage which would render the projectile unsuitable for its intended use.

3.3.2.1 Transportation vibration. The containerized projectile on a pallet shall withstand the random vibration in accordance with Figures 1-6.

3.3.2.2 Loose cargo transportation. The containerized projectile, when resting on a horizontal surface, shall withstand exposure to 1 inch double amplitude, circular path motion of the surface in a vertical plane at 284 $\pm$ 2 revolutions per minute.

3.3.2.3 Launch shock. The projectile shall withstand exposure to cannon launch accelerations at the following temperatures and levels:

Temperature	Maximum Launch Shock (forward g)
-25°F (-31.7°C)	6800
70°F (21°C)	7500
145°F (62.8°C)	8100.

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3.4 First article inspection. This specification makes provisions for first article inspection. First article samples shall meet all the requirements of this specification. Requirements for submission of the first article samples by the contractor shall be as specified in the contract.

3.5 Workmanship. All parts shall be fabricated and finished in a thoroughly workmanlike manner. Except as otherwise specified, the parts shall be free of cracks, scratches, burrs, dents, gouges, surface defects or any surface discontinuities. The items shall be free from dirt, chips, grease, corrosion and other foreign material. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and legible. Exterior surface coatings shall be continuous. Scratches not exposing base metal are permitted.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the procuring activity. The procuring activity reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item.

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of 20 projectiles and the parts and assemblies listed below.

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All assemblies and components shall have been produced by the contractor or furnished by a supplier and shall have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to ensure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings for materials, processes, and tests. A sample containing known defects will not be submitted unless specifically authorized by the procuring activity.

PART DESCRIPTION	DRAWING	QUANTITY
Ring, Retaining	9305330	5
Stud, Threaded	9305316	5
Ratchet, Machining	9305319	5
Bolt, Flat, 100° reduced head, hi torque, special	9305326	5

4.3.2 Inspections to be performed. First article test samples may be subjected by the procuring activity to any or all of the examinations and tests specified in this specification and to any or all requirements of the applicable drawings. Environmental tests shall be conducted in the sequence indicated. The cannon launch test conditions shall be as defined for rounds 1 through 20 in Table II. The testing shall be considered destructive and the tested assemblies shall not be returned to a lot for delivery. Subsequent to environmental testing of the containerized projectile, and prior to cannon launch, the projectile shall meet the requirements of 4.5.6 or the first article sample shall be rejected.

4.3.3 Rejection. If any test sample fails to comply with any of the applicable requirements, the first article samples shall be rejected. When launched, not more than 2 projectiles in each temperature group of 10 projectiles shall fail to function as defined in 3.3.1.3 or the first article sample shall be rejected. A no-test shall be declared if any of the conditions specified in 4.5.1.5 occur. The procuring activity reserves the right to terminate its inspection upon any failure of any test sample to comply with any of the stated requirements. Corrective actions to be performed by the contractor on first article test samples shall be approved by the procuring activity prior to implementation.

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TABLE II. Projectile test firing conditions

Round No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Range/Trajectory																				
12 km Glide	X	X			X	X			X	X				X	X	X	X			
4 km Ballistic			X	X			X	X			X	X	X					X	X	X
Temp Condition																				
Hot	X	X	X	X						X	X					X	X	X	X	
Cold					X	X	X	X	X			X	X	X	X					X

Range/Target Conditions

Target - Stationary steel plate, a minimum of 1 inch thick, 10 feet by 10 feet minimum size, oriented within ± 10 degrees of normal to the predicted projectile velocity vector.

Target reflectivity - Sufficient to insure a minimum of 7 millijoules of laser energy reflected in the direction of the projectile approach path.

Atmospheric visibility - Greater than 10 km at the target.

Cloud ceiling - Greater than 3 km over the projectile trajectory.

Wind - Less than 25 knots over the projectile trajectory.

Precipitation - None over the projectile trajectory.

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TABLE III
FIRST ARTICLE INSPECTIONS, PROJECTILE

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63223A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9305300
	PROJECTILE, GUIDED M712				NEXT HIGHER ASSEMBLY Not Applicable
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Projectile with tactical warhead section (Note 1)				
Critical					
1	Warhead integrity	20	-	3.3.1.2	4.5.4
Major					
101	Transportation vibration (Note 2)	20	-	3.3.2.1	4.5.5.1
102	Loose cargo transportation (Note 2)	20	-	3.3.2.2	4.5.5.2
103	Post environmental test inspection	20	-	3.3.2	4.5.6
104	Launch shock (flight test)	20	-	3.3.2.3	4.5.7
Minor					
201	Workmanship	20	-	3.5	Visual

NOTES:
1 - No deviation from sequential order shown above for Major items 101 through 104.
2 - Test performed with projectile installed in shipping container (drawing 9300440).

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot. Unless otherwise specified herein or by the procuing activity, inspection lot size, lot formation, and presentation of lots shall be in accordance with the inspection and lot sampling procedures herein and the "submission of product" provisions of MIL-STD-105. Lot size shall be not greater than 2 month's production or 500 projectiles which ever is less. Each lot of projectiles shall consist of warhead sections, control sections and guidance sections from not more than one interfix number from one manufacturer.

4.4.2 Inspections.

a. Sampling plans. Unless otherwise specified, sampling shall be to level II of MIL-STD-105.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63223A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9305330
4.4.2.1	RING RETAINING				NEXT HIGHER ASSEMBLY 9305300
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Width		0.40%	Dwg	Gage
102	Thickness		0.40%	Dwg	Gage
103	Outside diameter		0.40%	Dwg	Gage
104	Raw material		N/A	Dwg	Vendor lot certification
105	Hardness after heat treat		N/A	Dwg	Vendor lot certification
Minor					
201	Workmanship		0.65%	3.5	Visual
202	True position of gap with respect to O.D.		0.65%	Dwg	Gage

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63223A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9305316
4.4.2.2	STUD, THREADED				NEXT HIGHER ASSEMBLY 9305300
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Thread dimensions		0.65%	Dwg	Gage
202	Hardness after heat treat		0.65%	Dwg	Gage
203	Workmanship		0.65%	3.5	Visual
204	True position of slot with respect to datum		0.65%	Dwg	Gage

NOTES:

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63223A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9305319
4.4.2.3	RATCHET, MACHINING				NEXT HIGHER ASSEMBLY 9305300
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	True position of thread minor diameter with respect to O.D.		0.65%	Dwg	Gage
202	Thread dimensions		0.65%	Dwg	Gage
203	Workmanship		0.65%	3.5	Visual

NOTES:

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63223A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9305326
4.4.2.4	BOLT, FLAT, 100° REDUCED HEAD, HI TORQUE, SPECIAL				NEXT HIGHER ASSEMBLY 9305300
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Material		N/A	Dwg	Vendor lot certification
202	Workmanship		.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.5	PROJECTILE, GUIDED M712 (See note 1)				9305300
					NEXT HIGHER ASSEMBLY
					Not Applicable
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Warhead Integrity		4.4.3.1	3.3.1.2	4.5.4
Major					
101	Warhead section/control section cable resistance		100%	3.3.1.1	4.5.3
102	Weight		0.40%	Dwg	Gage
103	Harness workmanship		0.40%	3.3.1	9300940, Visual
104	Guidance section/warhead section coupling screw torque		0.40%	Dwg	Gage
105	Warhead section/control section ratchet torque		100%	Dwg	Gage
106	Center of gravity		0.40%	Dwg	Gage
107	Projectile straightness		100%	Dwg	Gage
108	Clearance between stud and control section and warhead section		100%	Dwg	Gage
109	Painting of dome (window)		100%	Dwg	Gage/visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9305300
4.4.2.5	PROJECTILE, GUIDED M712 (See note 1)				NEXT HIGHER ASSEMBLY Not Applicable
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
110	Torque and sealing compound per note 2		0.40%	Dwg	Gage/visual
111	Torque and sealing compound per note 7		0.40%	Dwg	Gage/visual
112	Transportation vibration (Note 2)		4.4.3.1	3.3.2.1	4.5.5.1
113	Loose cargo transportation		4.4.3.1	3.3.2.2	4.5.5.2
114	Post environmental test inspection		4.4.3.1	3.2.2	4.5.6
115	Launch shock (flight test)		4.4.3.1	3.3.2.3	4.5.7
Minor					
201	Projectile marking		0.65%	Dwg	Visual
202	Thermal compression bonding		Dwg	Dwg	9300940
203	Adhesive bond cleaning		Dwg	Dwg	9300975
204	Application of sealing compound		Dwg	Dwg	9300994
205	Workmanship		0.65%	3.5	Visual
206	Packaging		0.65%	5.0	Visual
207	Overall length		0.65%	Dwg	Gage
208	Fin operation		0.65%	Dwg	Gage

NOTES:
1 - Major items 112 thru 115 shall be conducted in the order indicated.
2 - Do not remove projectile from container between testing of major items 112 and 113.

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4.4.3 Testing.

4.4.3.1 Sequential sampling for flight testing. Each inspection lot shall be subjected to the flight test by the procuring activity in accordance with paragraph 4.5.7. A sample of 30 projectiles from each inspection lot shall be selected at random by the procuring activity for the flight test. The test samples shall be selected from the production run throughout the period that an inspection lot is being produced. After selection, the test sample shall be stored in containers (MIL-P-63227) in a bonded area and no further tests on the sample shall be performed by the contractor. The acceptance/rejection criteria shall be in accordance with the sequential sample plan of Table V. In the event that an accept/reject decision is determined before all 30 projectiles have been flight tested, the remaining projectiles shall be returned to the lot for delivery. The environmental test performed on the projectile is not considered destructive and projectiles that have been subjected to environments test and not flight tested shall be returned to the lot for delivery. The cannon launch test conditions shall be in accordance with Table IV.

4.4.3.2 Soft recovery and live sequence test. Soft recovery and live sequence tests shall be conducted on a sample of three (3) projectiles (or sets of sections) per month for information only. The information expected to be obtained from this test is the following:

a. An assessment of the reliability of the projectile under the worst case environments.
b. Identify root causes of failures induced by the environmental conditions and launch acceleration.
c. Achieve a look ahead capability to uncover failure modes as soon as practical.

The contractor shall select the three (3) projectiles from the beginning of the month. The projectiles shall be subjected to the tests in accordance with 4.5.8. The contractor shall submit to the procuring activity a report of results obtained from these tests.

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TABLE IV. Launch acceleration and temperature distribution for LAT

A

TEMPERATURE	LAUNCH ACCELERATION	
	LOW G (M3A1-Z4,Z5,M4A2-Z6)	HIGH G (M119A2)
COLD (-25°F)	1, 10	4
AMBIENT	2,5,8,9	3,12
HOT (+145°F)	6, 11	7

B

TEMPERATURE	LAUNCH ACCELERATION	
	LOW G (M3A1-Z4,Z5,M4A2-Z6)	HIGH G (M119A2)
COLD (-25°F)	5, 10	1
AMBIENT	2,3,8,9	4.12
HOT (+145°F)	6, 11	7

When the previous LAT included a high G, cold round, the current LAT starts with subtable "A". When the previous LAT did not include a high G, cold round the current LAT starts with subtable "B". If the current LAT requires more than a group of twelve rounds, the firing for the second group shall use the opposite subtable.

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TABLE V. Sequential sample plan and accept/reject criteria

Number Samples	Number Failures	
	Accept	Reject
1	-	-
2	-	-
3	0	-
4	-	-
5	-	-
6	1	6
7	-	6
8	-	7
9	2	7
10	-	8
11	-	8
12	3	9
13	-	9
14	4	10
15	5	10
16	-	10
17	-	11
18	6	11
19	-	11
20	7	12
21	-	12
22	8	12
23	-	12
24	9	12
25	-	12
26	-	12
27	10	12
28	-	12
29	-	12
30	11	12

NOTE: The lot is rejected if any projectile detonates prior to fuze arming.

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4.5 Methods of inspection.

4.5.1 Test conditions.

4.5.1.1 Environmental conditions. Unless otherwise specified for the purpose of test, testing shall be conducted at the following conditions:

(a) Temperature: $72^{\circ} \pm 18^{\circ}F$ ($22.2^{\circ} \pm 10^{\circ}C$)

(b) Relative Humidity: 90 percent maximum

(c) Barometric Pressure: 28 to 32 inches of mercury

4.5.1.2 Test tolerances. All tolerances specified herein are absolute with no allowance made for measuring equipment inaccuracy. Measurements shall ensure that the values measured are within the tolerances of limits specified.

4.5.1.3 Inspection equipment. The inspection equipments required to perform the examinations and tests prescribed herein are described in the "Paragraph Reference/Inspection Method" column in the inspection tables incluced herein. The contractor shall submit for approval all inspection equipment designs in accordance with the terms of the contract (See 6.4).

4.5.1.4 Functional projectile determination. Any projectile component, parameter, or characteristic, found by inspection or test to be in a state that would cause the projectile to fail to function properly when cannon launched, shall be classified a projectile failure. Disposition of failed projectiles shall be the responsibility of the procuring activity. A functional projectile shall be a projectile which reaches the target area and intersects within a circle, in a plane perpendicular to the projectile velocity vector, having a radius of 5.0 feet from the designator spot centroid, and which detonates with a high order explosion under impact. The designator spot centroid is defined as the centroid of laser spot energy averaged over the last two seconds of projectile flight. The 3σ variation of the spot center during these last 2 seconds shall be held to a one foot diameter circle.

4.5.1.5 "NO-TEST" determination. A no-test shall be declared if any of the following conditions occur:

(a) A failure to function properly caused by a malfunction of interfacing equipment of systems.

(b) Failure of Government Furnished Property (GFP) to perform within its specification requirements.

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(c) A malfunction resulting from test item abuse, testing beyond specification limits and conditions, or failure to follow prescribed operational procedures.

(d) An "act of God" condition causing a failure.

4.5.2 Physical and dimensional characteristics inspection. The physical and dimensional characteristics shall be inspected to the applicable requirements of drawing 9305300.

4.5.3 Warhead section/control section cable resistance test. The Warhead Section/Control section cable resistance shall be tested in accordance with and shall meet the requirements of 3.3.1.1. The testing shall observe the "CAUTION" of 3.3.1.1.

4.5.4 Warhead integrity test. The warhead section shall be subjected to radiographic inspection to verify the requirements of 3.3.1.2. Both safe-and-arm mechanisms shall be in the "SAFE" condition.

4.5.5 Environmental tests. Environmental tests shall be performed on the containerized projectile.

4.5.5.1 Transportation vibration. Six (6) containerized projectiles shall be palletized in accordance with drawing 19-48-4159-20PM-1003 when it is subjected to the transportation vibration test. Dummy projectiles which simulates the weight and center of gravity of the projectile may be used in the event that less than six (6) projectiles are to be tested. Each pallet of projectiles shall be subjected to the following vibration:

Condition	Time	
5 Ton Vertical	24 min	Figure 1
5 Ton Longitudinal	24 min	Figure 2
5 Ton Transverse	24 min	Figure 3
12 Ton Vertical	60 min	Figure 4
12 Ton Longitudinal	60 min	Figure 5
12 Ton Transverse	60 min	Figure 6

The vibration may be applied in either each axis separately or all three axis simultaneously. Positive control of input energy levels shall be maintained and monitored (closed loop test).

4.5.5.2 Loose cargo transportation. The containerized projectile shall be subjected to loose cargo transportation testing in accordance with MIL-STD-810, method 514.2 Procedure XI, Table 514.2-VII, with the exception that the total period of vibration shall be 30± 1.0 minutes (7.5 minutes/lateral side).

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4.5.6 Post environmental test inspections. After exposure to the environmental tests of 4.5.5, each projectile shall be removed from its container and shall be inspected for damage. Specific inspections shall include as a minimum:

(a) There shall be no loose screws or loose parts.

(b) The wings have remained locked in their folded positions. Unlatched fins shall not be cause for rejection. Fins shall be reinserted and latched in the retracted position.

(c) There shall be no cracks in the seeker window.

(d) The seeker shall be subjected to radiographic inspection in two planes 90° apart and normal to the longitudinal axis, and shall show no evidence of failure, such as disengagement of the gotcha lock.

(e) The warhead shall be subjected to radiographic inspection in accordance with 4.5.4

Any projectile found by inspection to be in a state that would cause the projectile to fail to function properly when cannon launched, shall be classified a failure. Disposition of failed projectiles shall be the responsibility of the procuring activity.

4.5.7 Flight test. Flight testing shall be conducted in accordance with TABLE II for first article test or TABLE IV for quality conformance inspections. Flight testing shall be conducted by the procuring activity after exposure to the environmental conditions of paragraph of 4.5.5. The laser designator shall have the characteristics of MIS 23152B. Laser codes shall be selected by the procuring activity from the Band II tri-service code assignments. Contianerized projectile conditioning shall be as follows.

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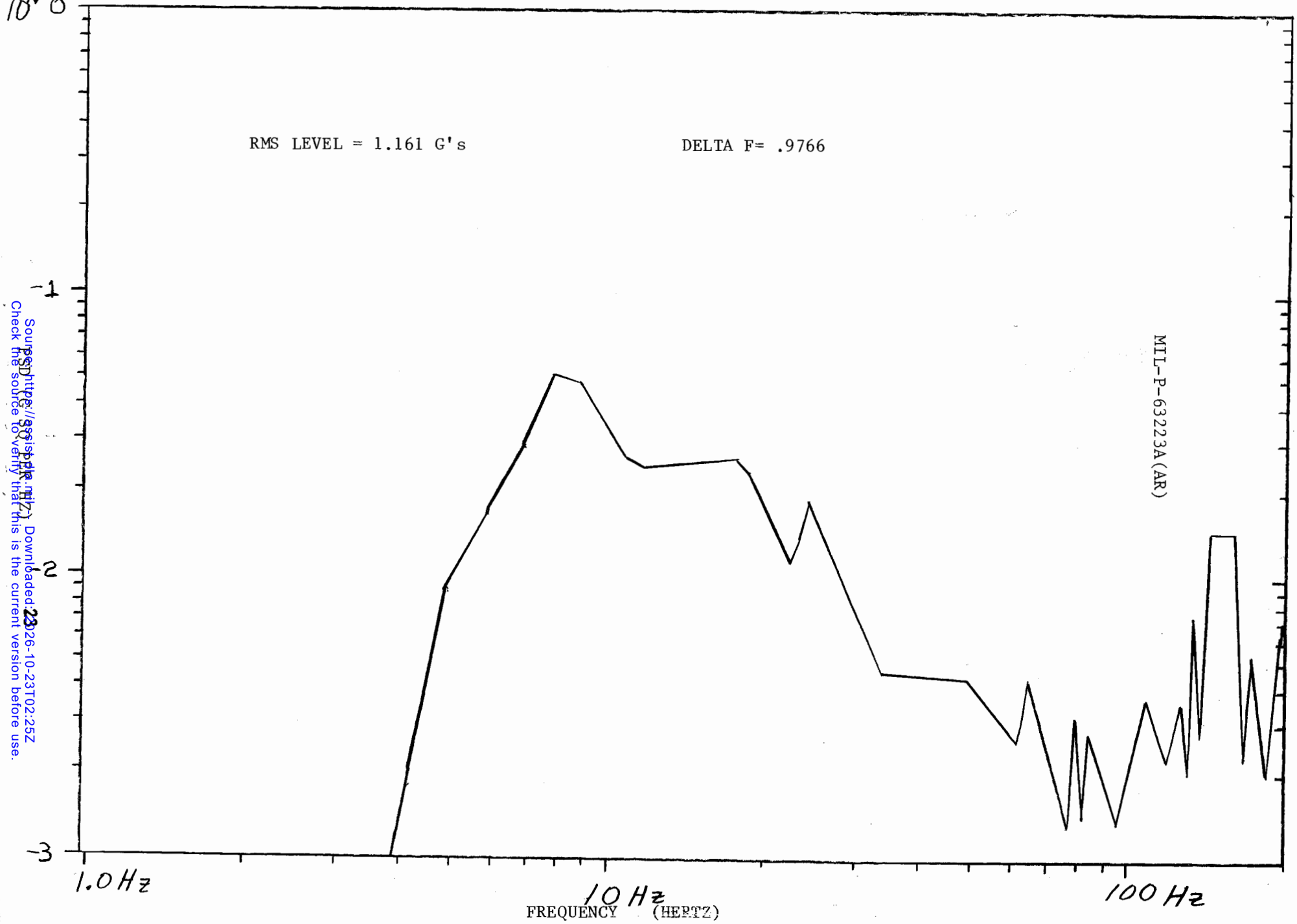


FIGURE 1. Break points describing the random vibration for 5 ton vertical.

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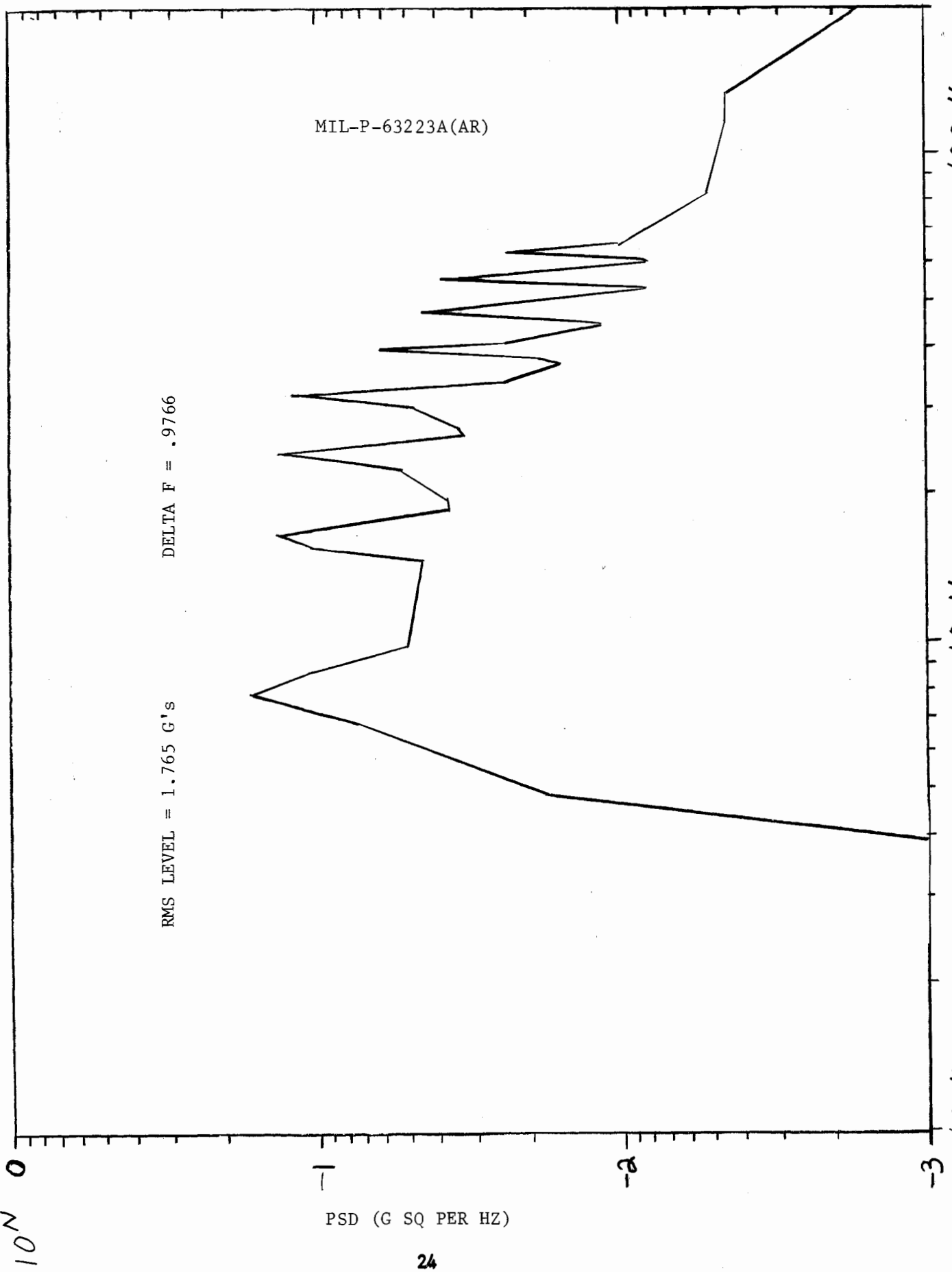


FIGURE 2. Break points describing the random vibration for 5 ton longitudinal.

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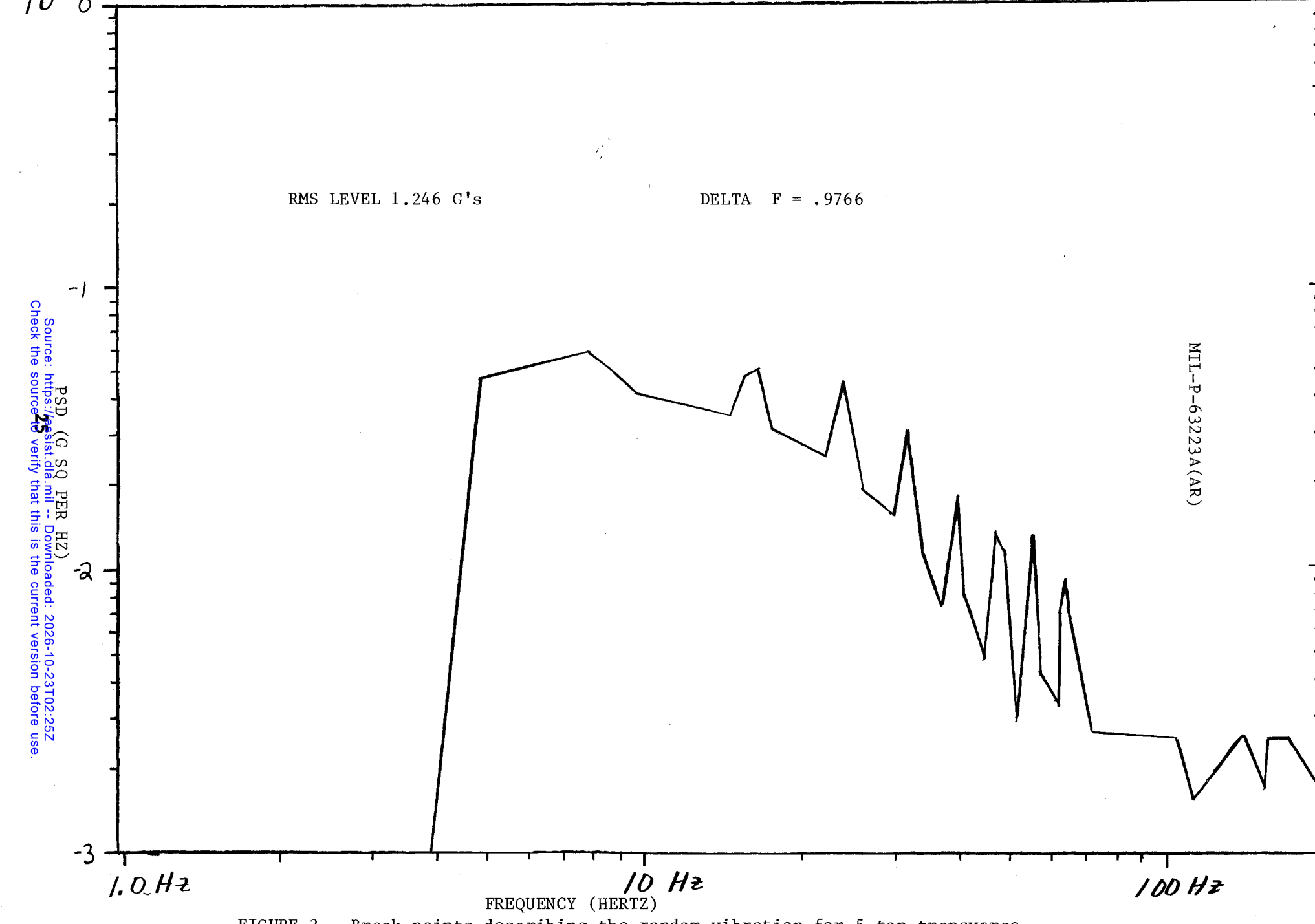


FIGURE 3. Break points describing the random vibration for 5 ton transverse.

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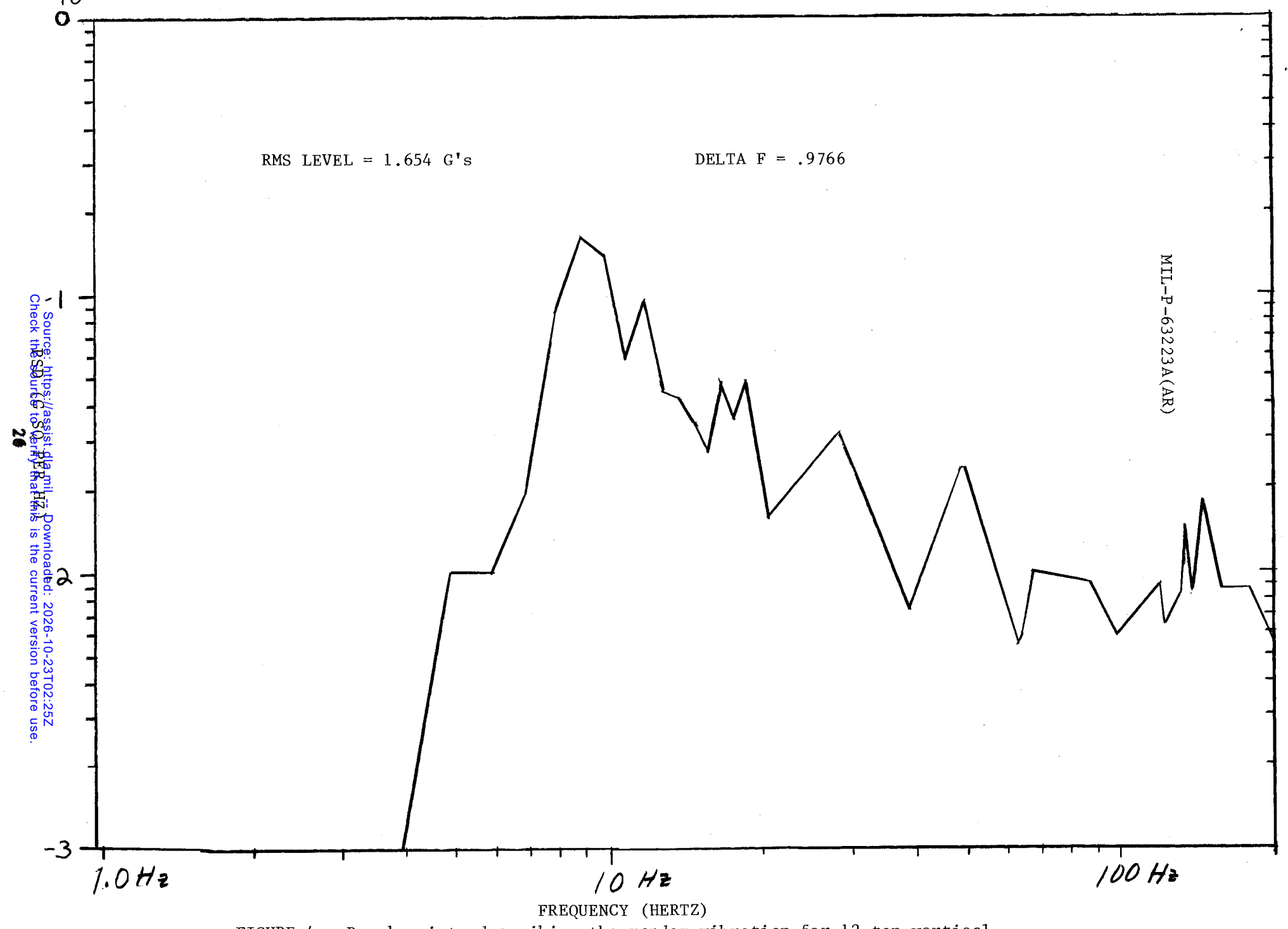


FIGURE 4. Break points describing the random vibration for 12 ton vertical.

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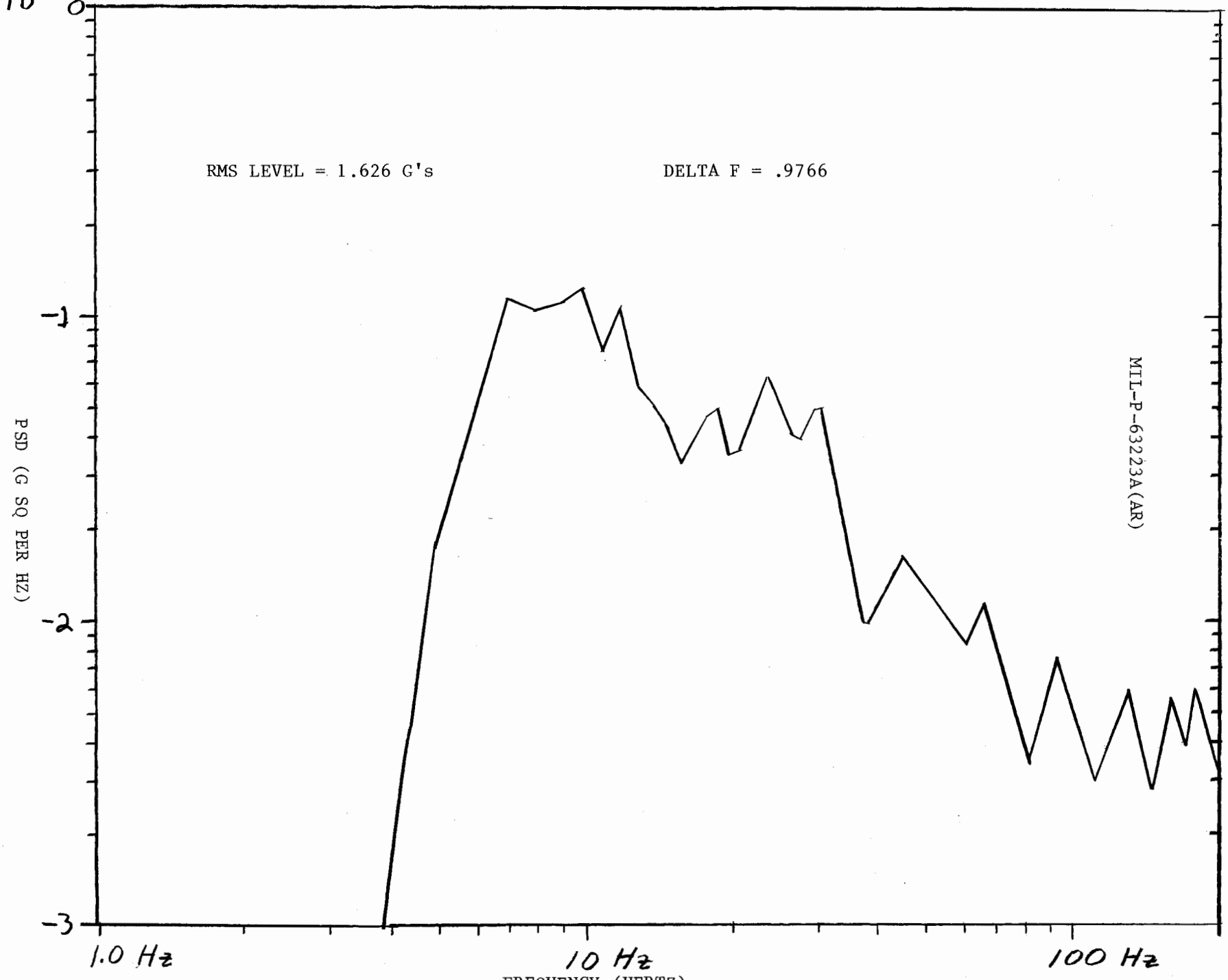


FIGURE 5. Break points describing the random vibration for 12 ton longitudinal.

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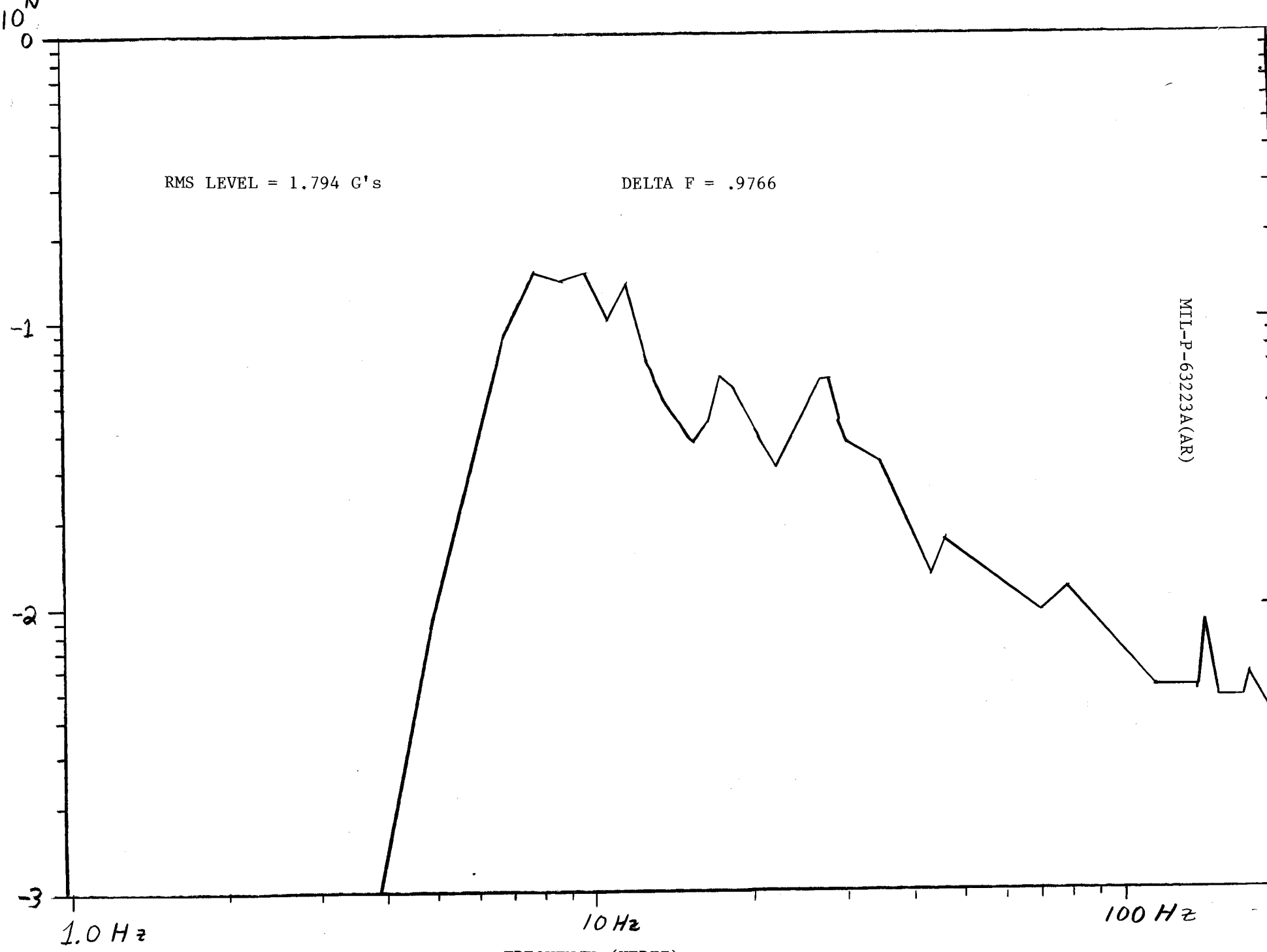


FIGURE 6. Break points describing the random vibration for 12 ton transverse.

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(a) Cold - The containerized projectiles shall be placed in an environmental chamber preconditioned to -25° ± 5°F (-31.7° ± 3°C) and maintained at that temperature for not less than 16 hours. At the conclusion of the conditioning period, and within 5 minutes after removal from the cold chamber, the projectiles shall be cannon launched with a charge that is also conditioned to the same temperature. If, during this procedure, extracton of a cold conditioned projectile is required for any reason, the projectile shall require conditioning to ambient temperature, drying, containerizing and environmental reconditioning to temperature prior to launch.

(b) Hot - The containerized projectiles shall be placed in an environmental chamber preconditioned to 145° ± 5°F (62.8° ± 3°C) and maintained at that temperature for not less than 16 hours. At the conclusion of the conditioning period, and within 5 minutes after removal from the hot chamber, the projectiles shall be launched with a charge that is also conditioned to the same temperature.

4.5.8 Soft recovery. After the projectile has been subjected to the environmental conditioning per 4.5.5.1 and 4.5.5.2 it shall be subjected to the live sequence test per 4.5.8.2 after it has been prepared and soft recovered per 4.5.8.1. The projectile shall be subjected to additional examinations and tests as required by the results of the live sequence test.

4.5.8.1. Parachute recovery. The seeker section or guidance section shall be removed and replaced by a parachute pack. The fins shall be removed and the +11 volt battery disconnected. The projectile shall then be cannon fired at high acceleration out of a 155MM cannon, using the M119A2 zone 8 at ambient temperature or equivalent propellant charge. The projectile and the seeker or guidance section shall be preconditioned at -25° ± 5°F as per 4.5.7(a) prior to cannon launch. The seeker section or guidance section shall be canister fired out of a 8 inch cannon using a propellant charge which will yield at least 6800 gs in acceleration.

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4.5.8.2 Live sequence test. This test is intended to simulate the function of the projectile as it would during flight. The live sequence test shall be conducted in accordance with paragraph 4.5.3.6.7 of MIL-P-63225 and paragraph 4.5.3.17 of MIL-P-63224.

5 PACKAGING

5.1 Packing, packaging, and marking. Packing and packaging of the projectile and marking of the loaded container shall be in accordance with drawing 9305335.

5.2 Palletization. Packaged projectiles shall be palletized for shipment in accordance with drawing 19-48-4159-20PM-1003.

6 NOTES

6.1 Intended use. The projectile is intended to be fired from a 155MM howitzer against a variety of military targets.

6.2 Ordering data. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification

(b) Exceptions, if any, to this specification

(c) Provisions for first article inspection

(d) Disposition of first article samples

(e) Requirements for lot control.

6.2.2 Contract data requirements.

(a) Contract data requirements for inspection equipment designs (conforming to data Item description DI-R-1714, tailored).

(b) Data Cards. When required by incorporation of DD form 1423 into the contract, data cards shall be prepared for each lot in accordance with Data Item description DI-L-1410, except that for propellant or explosive lots, Acceptance and Description Sheets will be prepared in accordance with MIL-STD-1171.

6.3 Definitions.

6.3.1 Contractor. Unless otherwise specified, the word contractor used throughout this specification shall mean any supplier or producer of items or material to the Government Owned, Contractor Operated (GOCO) plants.

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6.3.2 Technical data package (TDP). A technical description of an item adequate for use in procurement. The description defines the required design configuration and assures adequacy of item performance. It consists of all applicable technical data such as plans, drawings, and associated lists of specifications, standards, models, performance requirements, quality assurance provisions, and packaging data clarification of Quality Assurance terms and definitions are available in MIL-STD-109.

6.4 Inspection equipment designs. Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" heading in section 4, or on the equipment lists referenced on the equipment tabulation, whichever is specified in the item detail specification. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.4.2 and 6.4.3 should he elect to do so.

6.4.1 SIE designs. SIE designs may consist of any of the following:

(a) Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

(b) A source control drawing or a specification control drawing as defined in MIL-STD-100.

(c) An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When the envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.4.2 Contractor designs. Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered.

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6.4.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery scheduled as stipulated in the contract. All submissions shall be addressed to : Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: AMSMC-QAR-I (D), Dover, NJ 07801-5001.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as, US Army Armament Research and Development Command (ARRADCOM) or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to extensiveness of the changes.

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Preparing Activity:
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(Project 1320-A821)

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